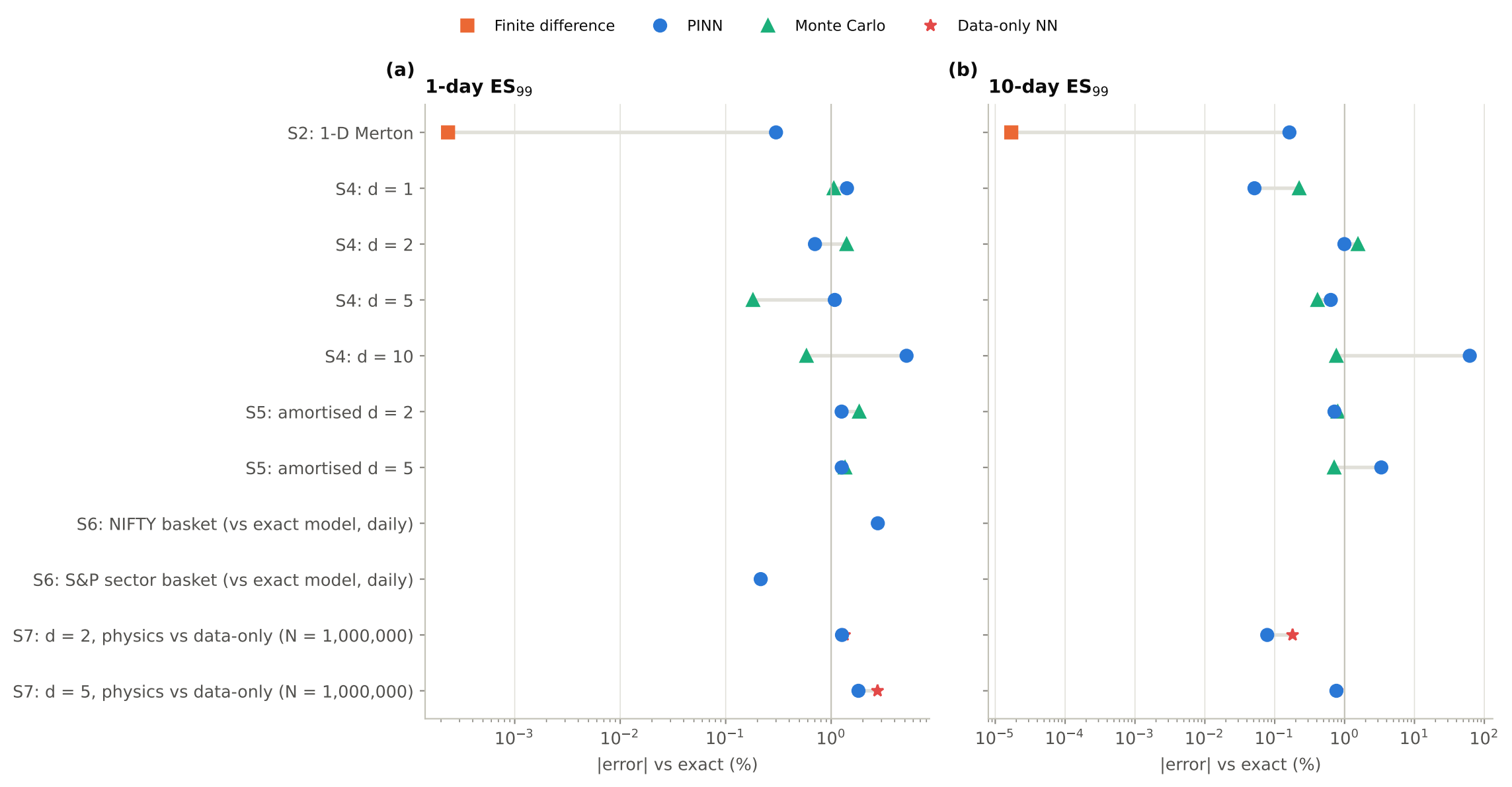
Finite difference
PINN
Monte Carlo
Data-only NN
(a)
1-day ES99
(b)
10-day ES99
S2: 1-D Merton
S4: d = 1
S4: d = 2
S4: d = 5
S4: d = 10
S5: amortised d = 2
S5: amortised d = 5
S6: NIFTY basket (vs exact model, daily)
S6: S&P sector basket (vs exact model, daily)
S7: d = 2, physics vs data-only (N = 1,000,000)
S7: d = 5, physics vs data-only (N = 1,000,000)
10−3
10−2
10−1
100
|error| vs exact (%)
10−5
10−4
10−3
10−2
10−1
100
101
102
|error| vs exact (%)